

Clustering-Based Pairs Trading

A from-scratch replication & extension of *Rotondi & Russo (2025)*

Headline: matched the paper's best strategy at **Sharpe 1.113** (paper: 1.01)
on a survivorship-bias-free 18-year in-sample backtest, validated on 5 years out-of-sample.

Built in Python on WRDS/CRSP data · ML clustering + statistical-arbitrage signal · full rolling-window backtest engine

The problem

Pairs trading = find two securities whose prices move together, trade the *spread* when it diverges, profit when it reverts.

The hard part is **pair selection** across a 1,000-stock universe:

- Brute-forcing all ~500,000 pairs is noisy and overfits.
- We want pairs that are *economically* related, not coincidental.

The paper's idea: use **unsupervised ML clustering** to group similar stocks first, then only form pairs *within* clusters. Fewer, higher-quality candidates.

Our goal: rebuild the whole pipeline from the data up, reproduce the published result independently, then extend it.

Data spine — built for honesty, not optics

Source	WRDS / CRSP daily (institutional-grade academic data)
Period	Jan 2000 – Dec 2025 (26 years)
Universe	991 stocks — share codes 10/11 only (US ordinary common)
Survivorship bias	Eliminated — delisted names kept; delisting returns modelled
In-sample	2003–2020 = 215 monthly returns
Out-of-sample	2021–2025 = 59 monthly returns (frozen, no re-tuning)

First 3 years are reserved purely as the formation window, so every reported return is genuinely out-of-sample. 79 names dropped vs the raw pull are fully accounted for (foreign / REITs / funds).

The pipeline (rolling, monthly)

```
CRSP returns → [1] DISTANCE metric → [2] OPTICS clustering
                |
                pairs only within a cluster
                ▼
[3] cointegration filter → [4] spread + rolling z-score
                |
                ▼
                [5] backtest: enter  $|z| > 2$ , exit  $z = 0$ 
```

Each month: form pairs on the **trailing 3 years**, trade them for the **next 1 month**, roll forward. 215 times (IS) + 59 times (OOS).

[1] Two distance metrics

How do we decide which stocks are "similar" before clustering?

- **SSD** — Sum of Squared Deviations between normalised price paths. Classic baseline (Gatev et al.).
- **PC — Partial-correlation distance** on market-adjusted returns. Strips out the common market factor, so it finds **idiosyncratic** co-movement — pairs that revert for stock-specific reasons. *The paper's winning metric.*

Distance feeds the clusterer; the metric choice turns out to be the single biggest driver of risk-adjusted performance.

[2] Clustering with OPTICS

OPTICS — density-based clustering (cousin of DBSCAN).

- No need to pre-specify the number of clusters.
- Leaves genuinely unrelated stocks **unclustered** instead of forcing them into a group.

Validation against ground truth (Dec 2023):

Metric	Ours (SSD)	Ours (PC)	Paper
# clusters	47	81	48 / 109
Purity vs SIC industry codes	0.871	0.937	0.81 / 0.84
GOOG & GOOGL co-clustered?	Y	Y	Y

xi hyperparameter tuned on cluster quality on 3 hold-out dates — **and frozen before ever looking at Sharpe**, to avoid look-ahead / overfitting bias.

[3] Cointegration filter & [4] the signal

Engle-Granger cointegration filter (*optional variant*) — keep a pair only if:

- spread rejects the unit-root null at **5%**, and
- spread **half-life in [5, 60] trading days** (fast enough to revert inside the window, slow enough to not be noise).

Trading signal — on the in-cluster, cointegrated pairs:

- Hedge ratio via OLS -> spread series
- **6-month rolling z-score** (strict look-ahead protection)
- **Enter** when $|z| > 2$ · **Exit** at $z = 0$ · force-close at month-end

Original extension: factor-beta clustering

A structurally different similarity metric — cluster by shared **risk-factor exposures**, not return co-movement.

- **Ridge-regress** each stock's returns on **18 factors** (6 FF style + 12 FF industry) → 18-d beta vector per stock
- **Distance** = standardised Euclidean between beta vectors

Metric	Sharpe	vs Paper
PC core	1.113	1.01 (match)
Factor-beta core	1.149	original extension, beats PC
Factor + coint filter	0.969	filter slightly reduces

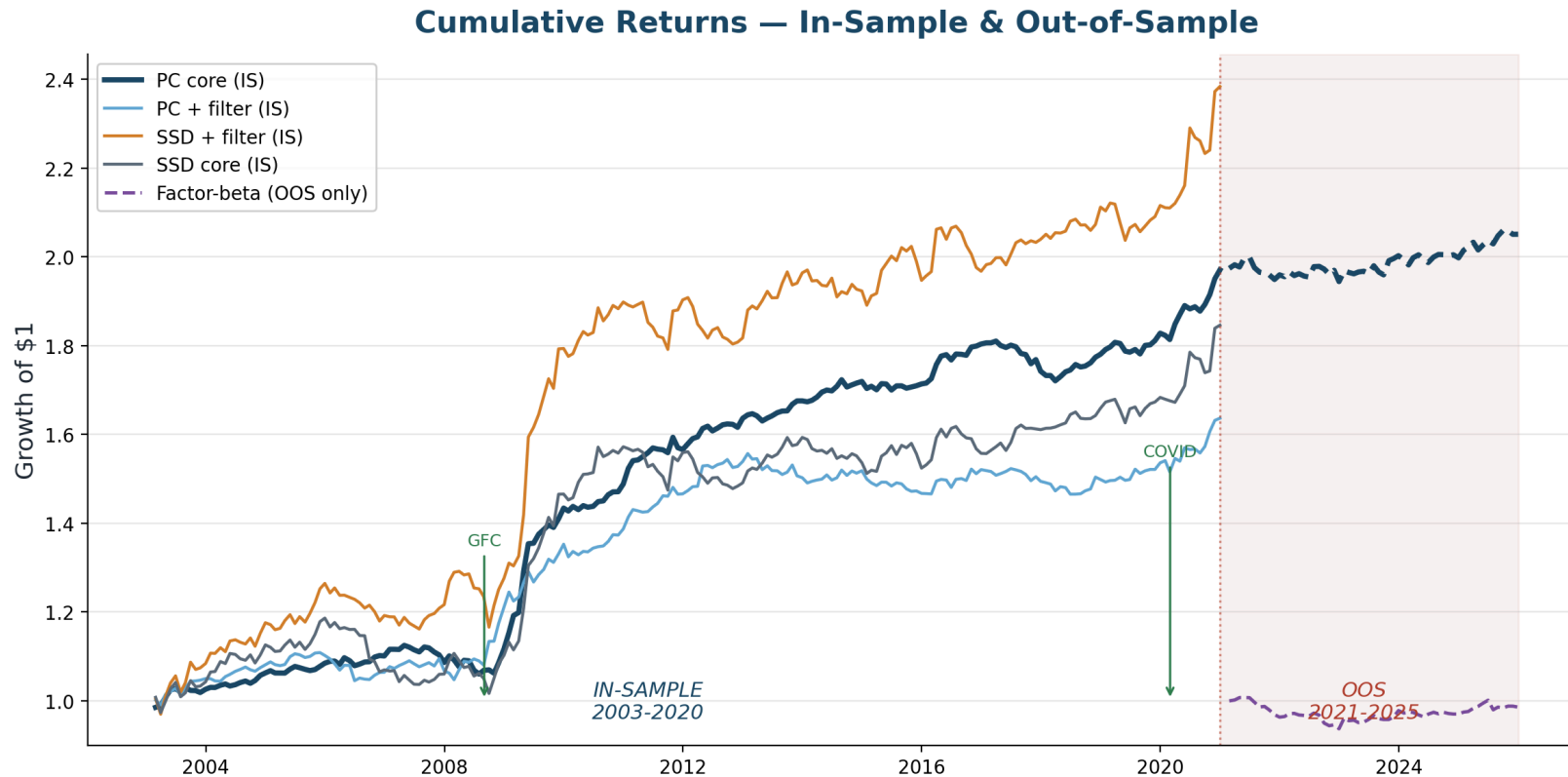
Two independent metrics reaching ~1.0 is strong evidence the edge is real, not a PC-specific artefact.

In-sample result (2003–2020)

Strategy	Ann. return	Ann. vol	Sharpe	Max DD
Factor-beta core	4.16%	3.61%	1.149	-3.59%
PC core	3.86%	3.46%	1.113	-5.75%
Factor + filter	5.16%	5.35%	0.969	-6.41%
SSD + filter	4.97%	6.37%	0.794	-9.74%
PC + filter	2.79%	3.57%	0.788	-5.88%
SSD core (<i>baseline</i>)	3.48%	5.54%	0.645	-14.31%

Factor-beta core (1.149) beats PC core (1.113) — two independent metrics above 1.1 is strong evidence the edge is real.

Cumulative returns — all strategies



PC core (dark) is the smoothest compounder. SSD strategies have higher absolute returns but far more volatility — risk-adjusted return is the metric that matters.

Why it works — diagnosed, not assumed

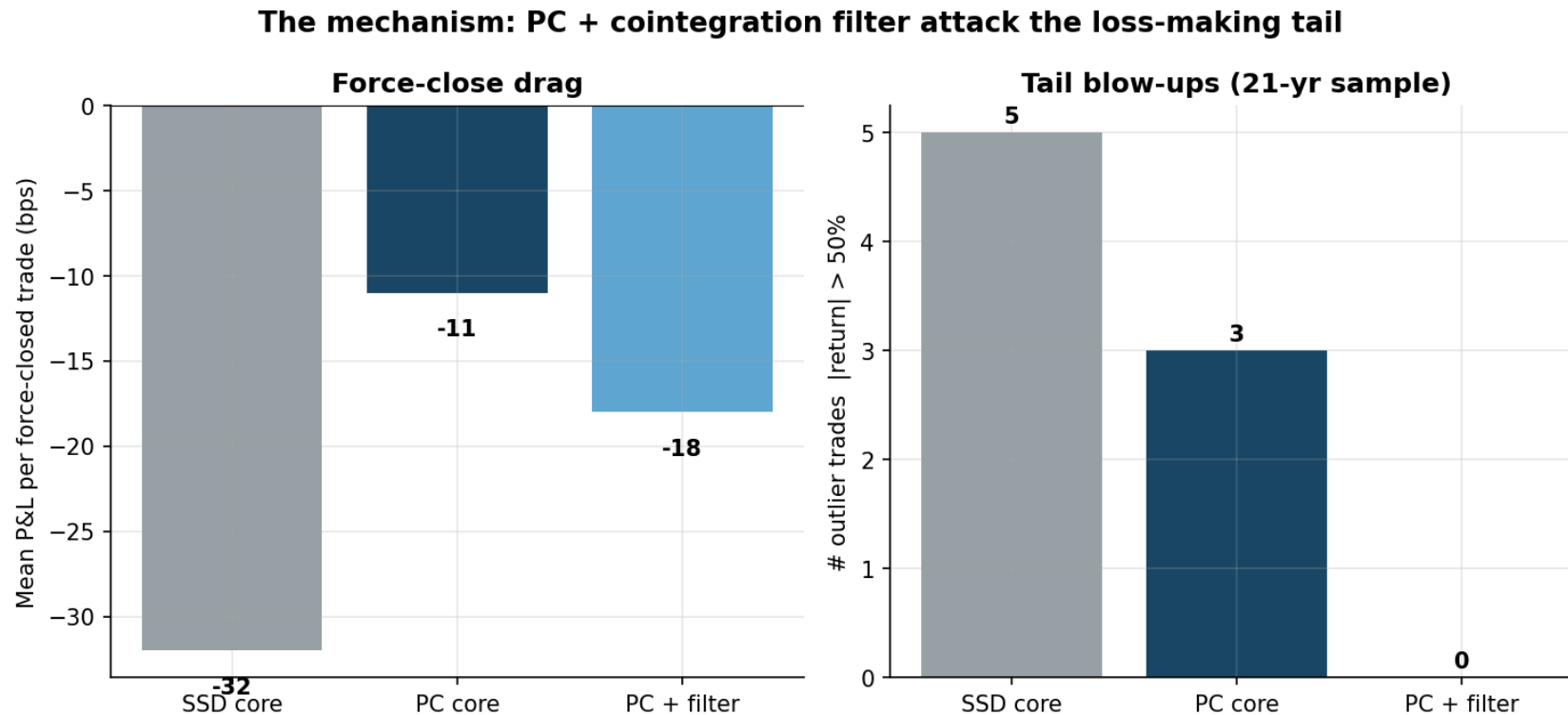
Before building Phase 2, we did a **P&L attribution** on the baseline and found a **bimodal** pattern:

- **11.4%** of trades cleanly revert: **+471 bps** each -> *all* the profit.
- **88.4%** get force-closed at month-end: **-32 bps** each -> a constant drag.

Thesis: the job isn't "find a higher Sharpe." It's **kill the force-close drag** by only trading pairs that actually revert.

This turned the next phase from guesswork into a targeted fix.

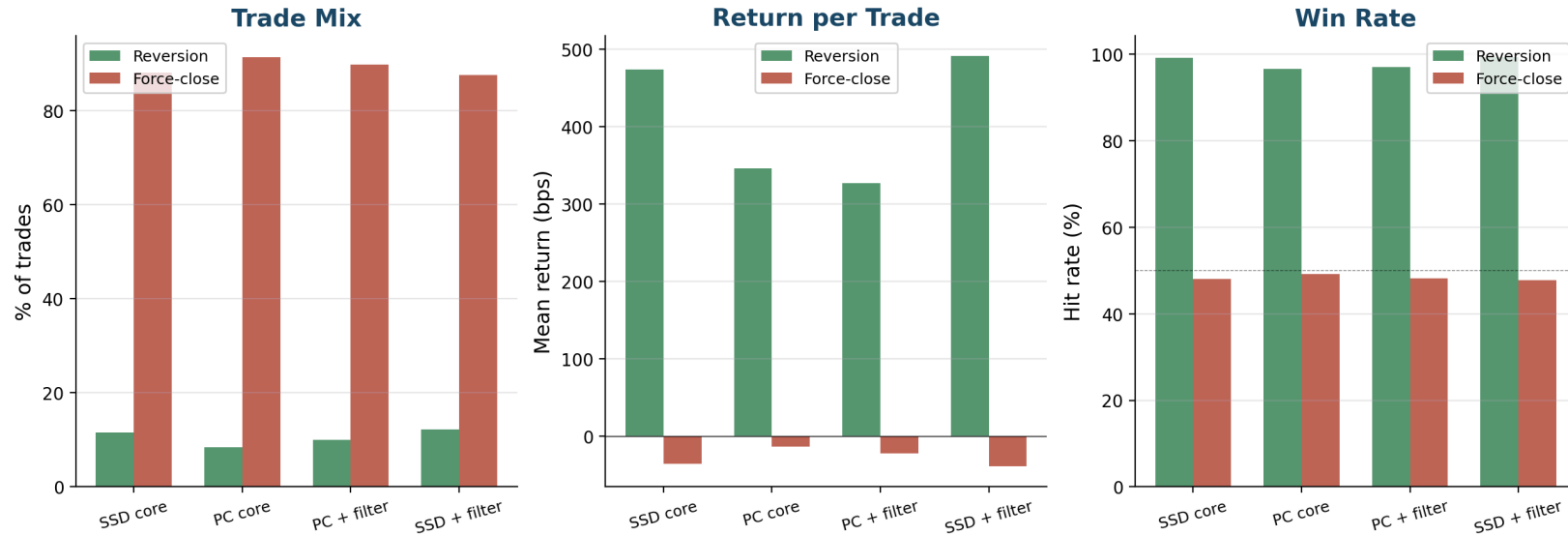
The mechanism — prediction confirmed



PC distance cut the per-trade force-close drag by **65%**; adding the cointegration filter **eliminated every tail blow-up** (0 trades with $|return| > 50\%$ in 18 years, vs 5 for the baseline).

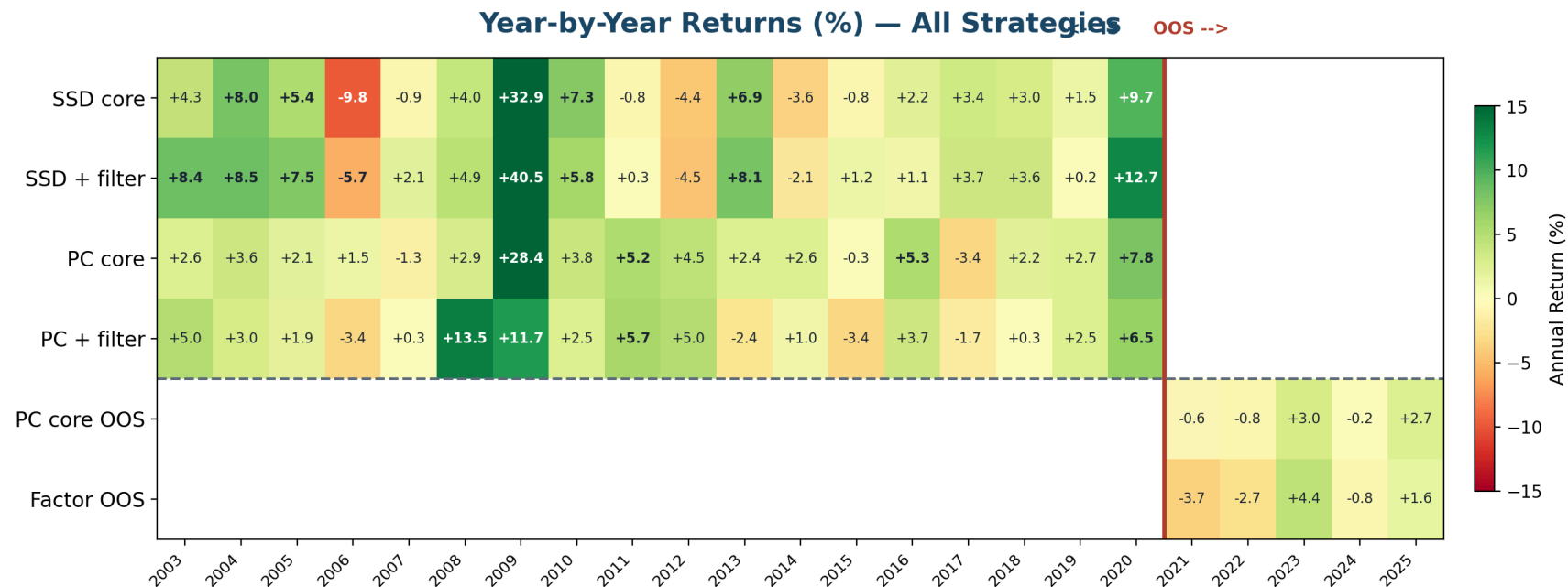
P&L breakdown by exit reason

P&L Breakdown by Exit Reason



Strategy	Reversion %	Reversion bps	Force-close %	Force-close bps
SSD core	11.6%	+474	88.1%	-36
PC core	8.4%	+346	91.4%	-14
PC + filter	10.0%	+327	89.7%	-23
SSD + filter	12.3%	+491	87.5%	-39

Year-by-year returns — regime dependence



Findings:

- Best year: 2009 (+28.4%) — GFC dislocation = highest dispersion, most reversions
- Calm periods (2013-19): avg +1.6%/yr — the strategy idles, doesn't lose
- OOS (2021-25): weak in low-dispersion 2021-22, recovered in 2023 & 2025
- 15/18 IS years positive (83%) — consistent but regime-dependent magnitude

Out-of-sample validation (2021–2025)

Strategy	IS Sharpe	OOS Sharpe	OOS Ann. Ret	OOS MDD	OOS Months
PC + filter	0.788	0.461	1.31%	-2.98%	59
PC core	1.113	0.412	0.82%	-2.75%	59
SSD + filter	0.794	0.224	0.82%	-5.52%	59
Factor + filter	0.969	0.131	0.42%	-6.56%	59
Factor core	1.149	-0.103	-0.29%	-6.99%	59

PC + filter is the best OOS strategy (0.461). The cointegration filter **helps both PC and factor** OOS — it removes pairs that drift apart. Filter rescues factor from -0.103 to 0.131.

The honest story: the strategy works, but it is regime-dependent. Performance tracks volatility and dispersion, not calendar time.

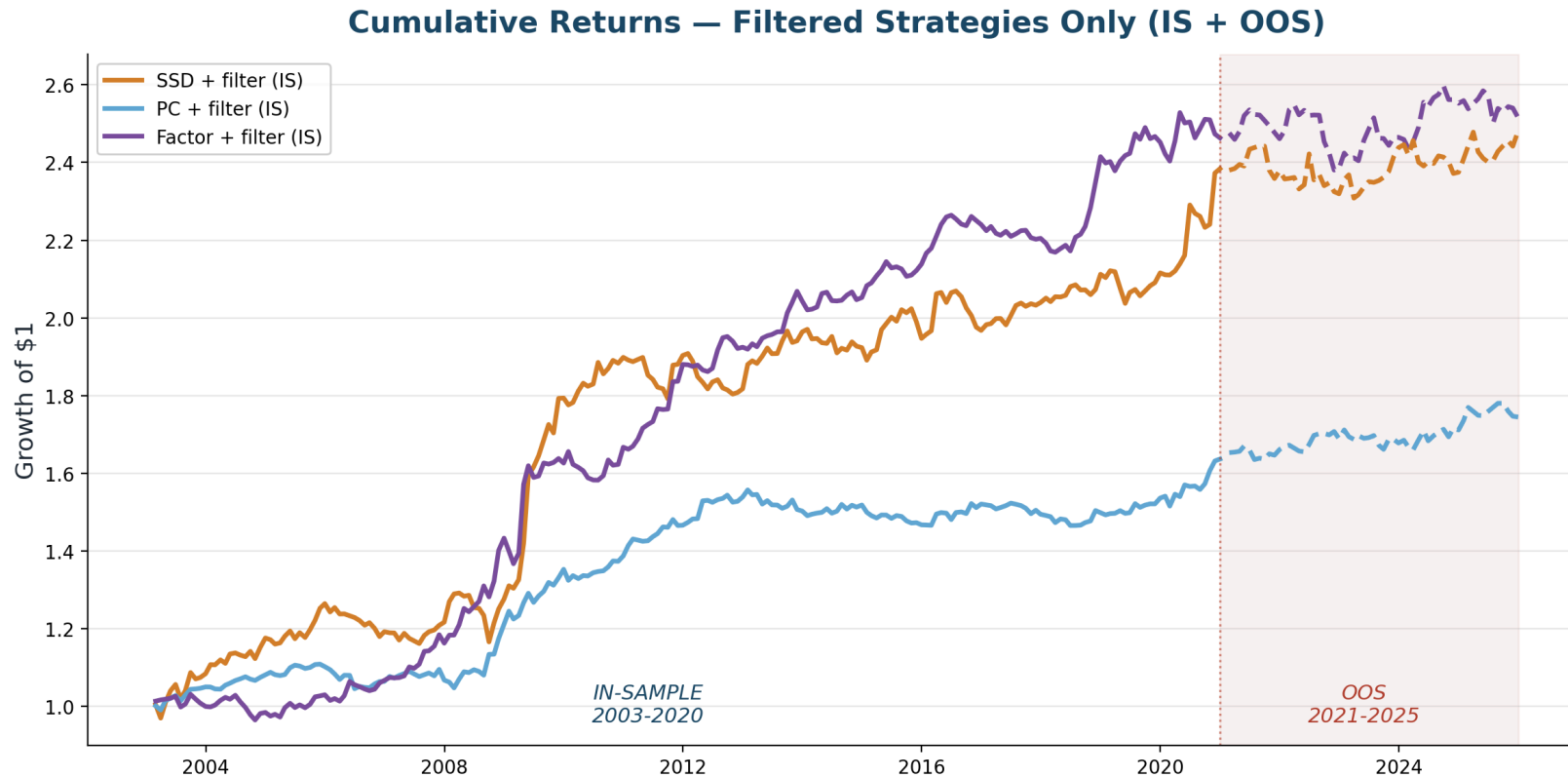
Filtered strategies — IS vs OOS scorecard

The cointegration filter is **required** by the project specification. Focus on filtered variants only:

Strategy	Period	Ann. Ret	Ann. Vol	Sharpe	Max DD	Months
Factor + filter	IS 2003–2020	5.16%	5.35%	0.969	-6.41%	215
SSD + filter	IS 2003–2020	4.97%	6.37%	0.794	-9.74%	215
PC + filter	IS 2003–2020	2.79%	3.57%	0.788	-5.88%	215
PC + filter	OOS 2021–2025	1.31%	2.92%	0.461	-2.98%	59
SSD + filter	OOS 2021–2025	0.82%	4.01%	0.224	-5.52%	59
Factor + filter	OOS 2021–2025	0.42%	3.77%	0.131	-6.56%	59

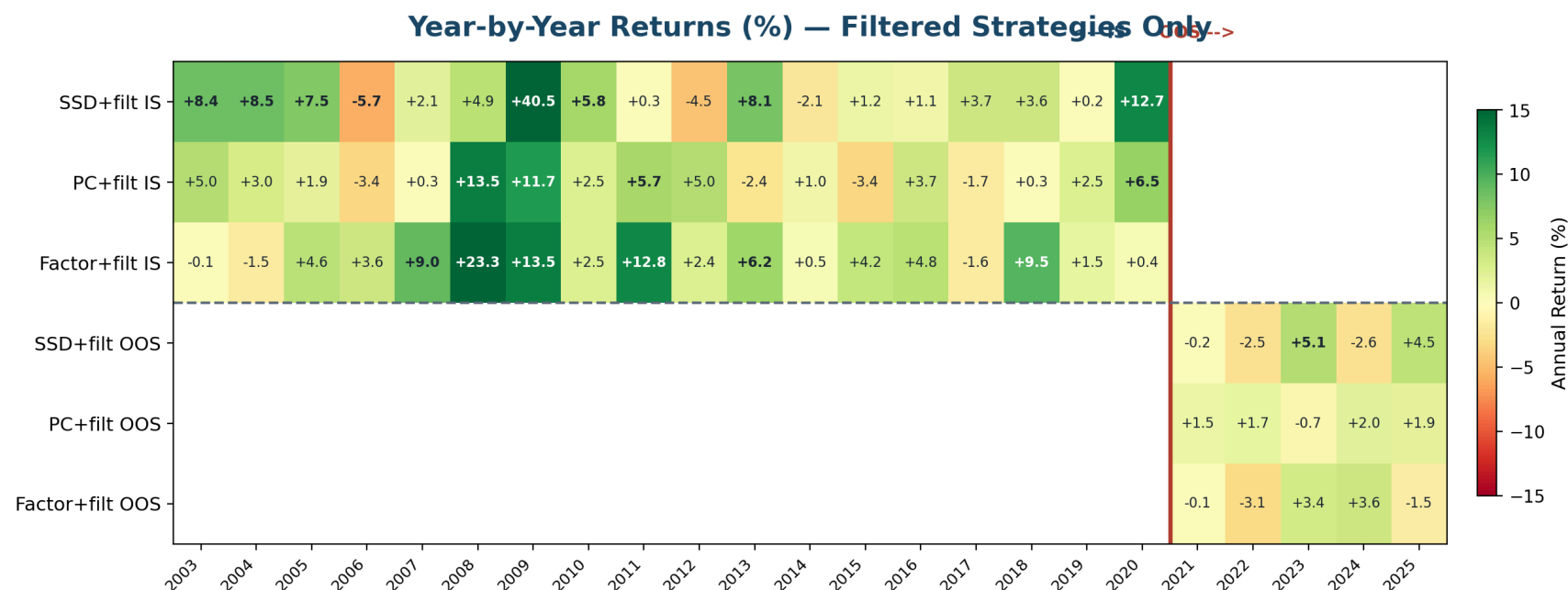
Key finding: PC + filter is the **best OOS filtered strategy** (0.461) — the filter actually *helps* PC in OOS. Filter rescues factor from -0.103 to 0.131.

Cumulative returns — filtered strategies (IS + OOS)



Solid = IS (2003–2020), dashed = OOS (2021–2025). All strategies use the Engle-Granger cointegration filter as required.

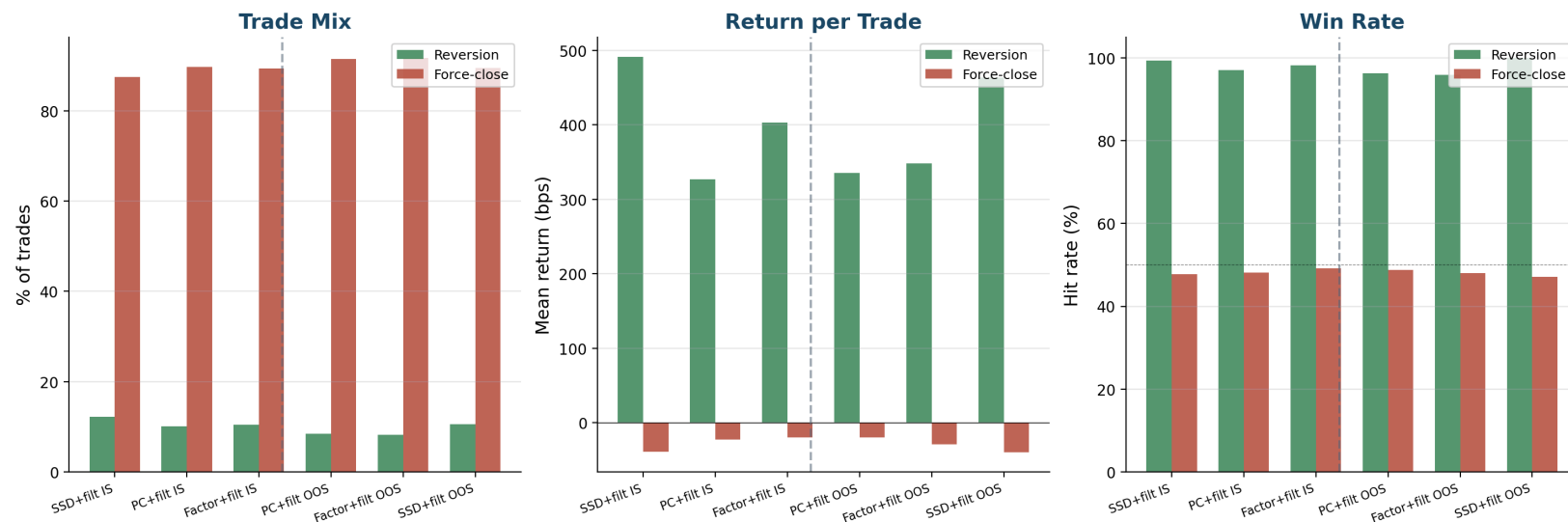
Year-by-year returns — filtered strategies



- **GFC (2008–09):** outsized returns for all filtered strategies — dislocation harvesting at its peak
- **Calm years (2013–19):** small but mostly positive — the filter keeps the strategy idle, not losing
- **OOS (2021–25):** positive years in 2023 & 2025 (high-dispersion); weak in low-dispersion 2021–22

P&L breakdown — filtered strategies (IS vs OOS)

P&L Breakdown — Filtered Strategies (IS vs OOS)



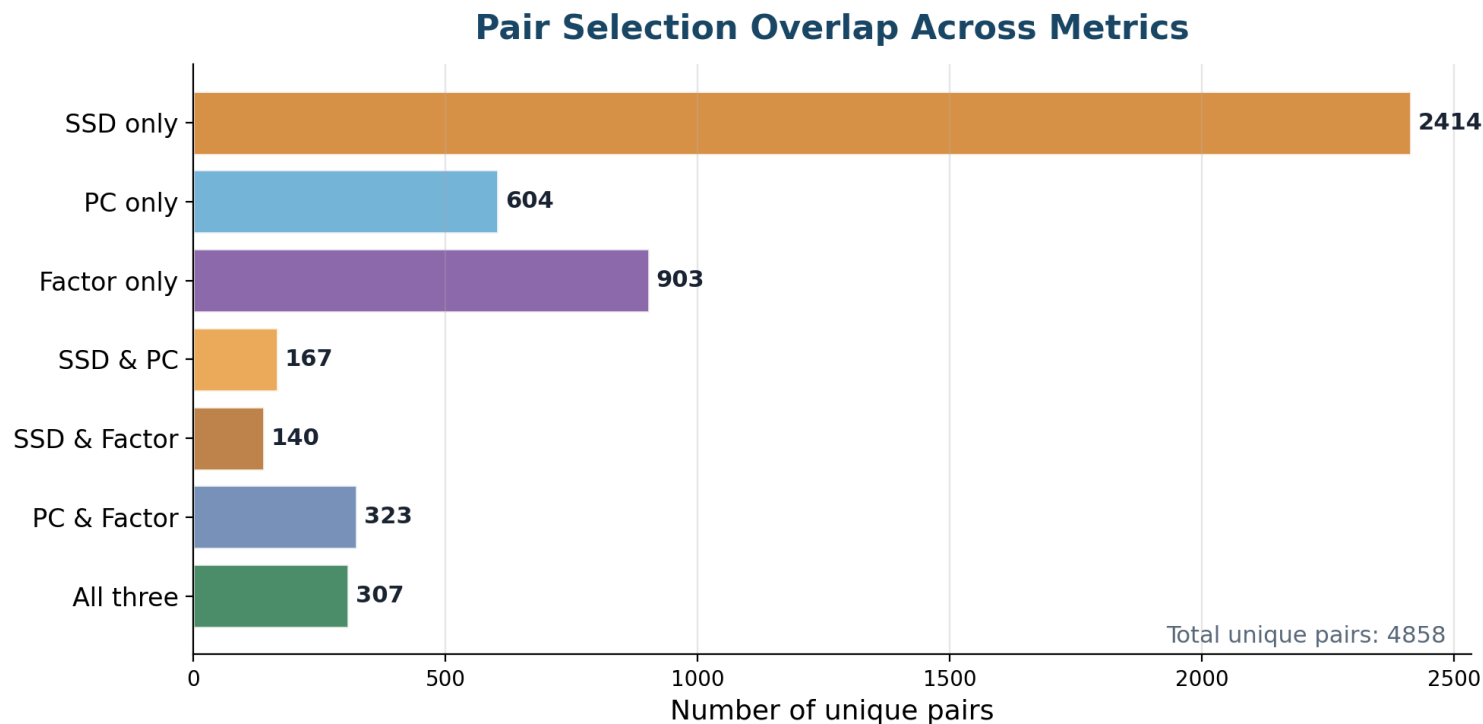
- **PC + filter:** lowest force-close drag among filtered variants
- **Factor + filter:** higher reversion rate but higher per-trade variance
- IS vs OOS comparison shows consistency of the trade mechanics across regimes

What each metric selects — example pairs

	SSD + filter	PC + filter	Factor + filter
#1 pair	CMS/XEL (utilities)	BK/NTRS (banks)	VZ/T (telecom)
#2 pair	LMT/RTN (defense)	VZ/T (telecom)	ED/SO (utilities)
#3 pair	RF/ZION (banks)	ED/SO (utilities)	WEC/XEL (utilities)
Unique to	MCO/UNH, JPM/GL	TMO/A, FDX/UPS	NKE/YUM, ORCL/QCOM
Total pairs	3,028	1,401	1,673
Total trades	8,143	7,594	4,871
Reversion %	12.3%	10.0%	10.5%

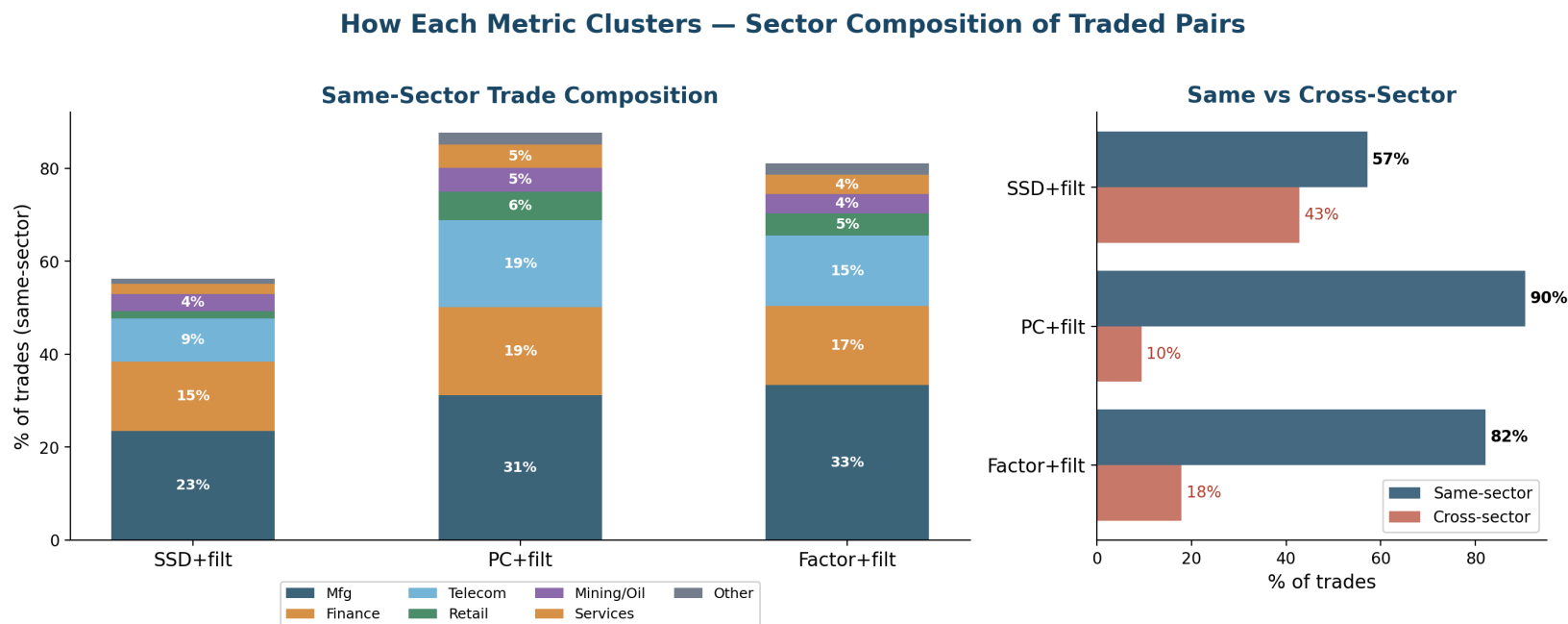
- **SSD** casts the widest net (3,028 pairs) — any stocks with similar price paths
- **PC** is the most selective (1,401 pairs) — only idiosyncratic co-movement survives
- **Factor** groups by risk exposure — finds cross-sector pairs like NKE/YUM (consumer beta)

Pair selection overlap across metrics



- **SSD: 2,414 pairs unique to it** — price-path similarity finds many candidates
- **PC: 604 unique pairs** — correlation-based filtering is the strictest
- **Factor: 903 unique pairs** — shared risk exposures find cross-sector relationships
- **Only 307 pairs found by all three** (of 4,095 total) — each metric sees different structure

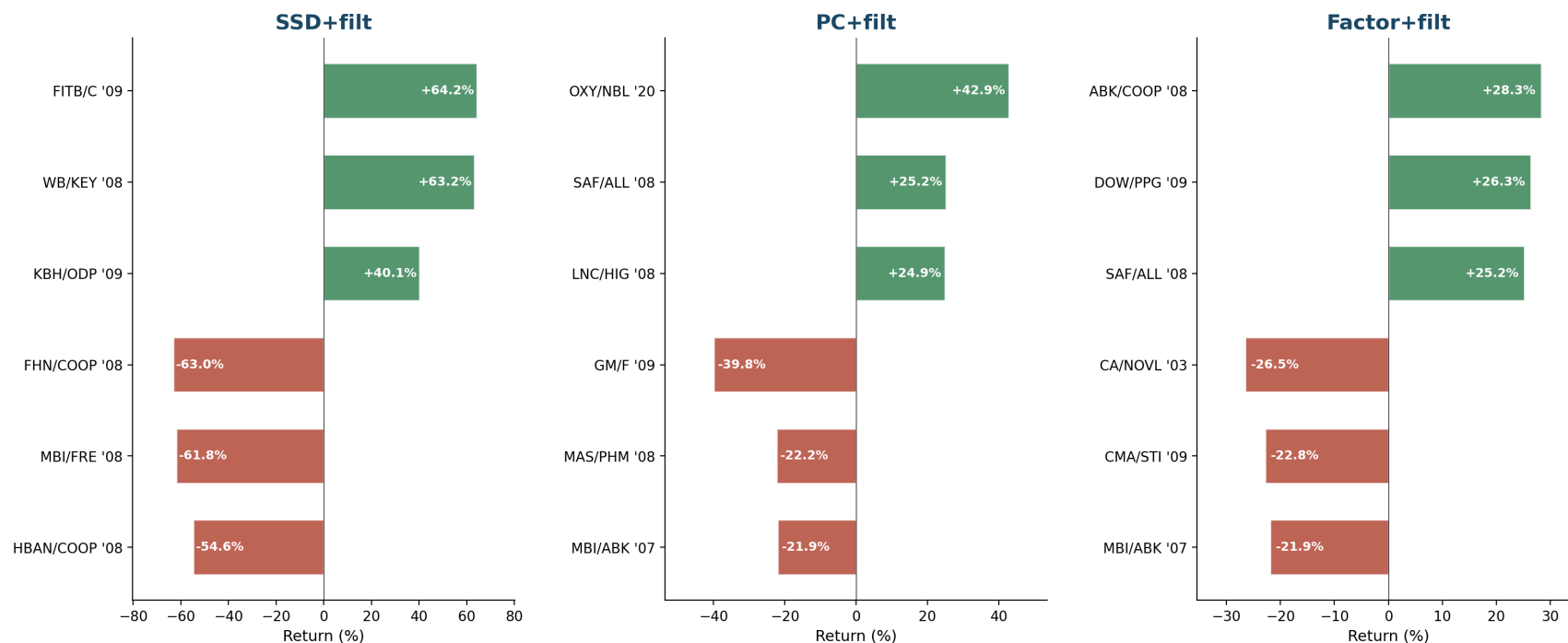
How each metric clusters — sector composition



- **PC is 91% same-sector** — idiosyncratic co-movement stays within industries (BK/NTRS, VZ/T)
- **SSD is only 57% same-sector** — price-path similarity crosses sector boundaries (XRX/LNC, HSY/NEE)
- **Factor is 82% same-sector** — shared risk factors mostly align with industry, but finds cross-sector pairs with similar beta profiles (NKE/YUM)
- PC's selectivity (fewer, tighter pairs) explains its lower volatility and smoother returns

Biggest winners & losers by metric (IS)

Top 3 Winners & Losers by Metric (IS, Filtered)



- **SSD** has the most extreme trades (+64% to -63%) — GFC bank pairs dominate both tails
- **PC** is more controlled (+43% to -40%) — GM/F (pre-bankruptcy) is the worst loss
- **Factor** is the most bounded (+28% to -27%) — fewer tail blowups
- Winners mostly revert; losers are almost all force-closes — confirming the bimodal thesis

Biggest winners & losers by metric (OOS)

	Winner	Return	Loser	Return
SSD	BIIB/CNC '21	+24.0%	FRC/EL '23	-61.2%
	CHTR/META '22	+20.6%	INTC/WDC '24	-18.6%
	SBUX/VRSN '24	+17.2%	VFC/WYNN '22	-18.3%
PC	EW/ZBH '24	+21.5%	GPC/LKQ '25	-13.2%
	DOW/CE '24	+12.6%	MU/WDC '25	-12.4%
	DOW/CE '25	+11.9%	FISV/FIS '23	-12.3%
Factor	BSX/CNC '21	+9.7%	PTC/IT '25	-19.4%
	WDC/QCOM '25	+9.6%	INTC/QCOM '21	-15.3%
	VFC/MHK '23	+9.2%	NVDA/MPWR '24	-14.9%

- SSD's biggest OOS loss: **FRC/EL (-61%)** — First Republic Bank collapse, Mar 2023
- **PC has the tightest loss range (-13%)** — idiosyncratic selection avoids blow-ups
- Factor's loss: **NVDA/MPWR** — AI momentum broke the factor-beta relationship

Carry-over: letting trades roll across months

Instead of force-closing every position at month-end, we allow trades to **roll over** (max 3 months).

- ~89% of trades in the baseline are force-closed — most haven't reverted yet
- Force-close drag is the #1 source of lost profit
- **Max hold: 3 months** (aligned with half-life upper bound of 60 trading days)
- Position still exits on reversion ($z=0$) or at the 3-month cap

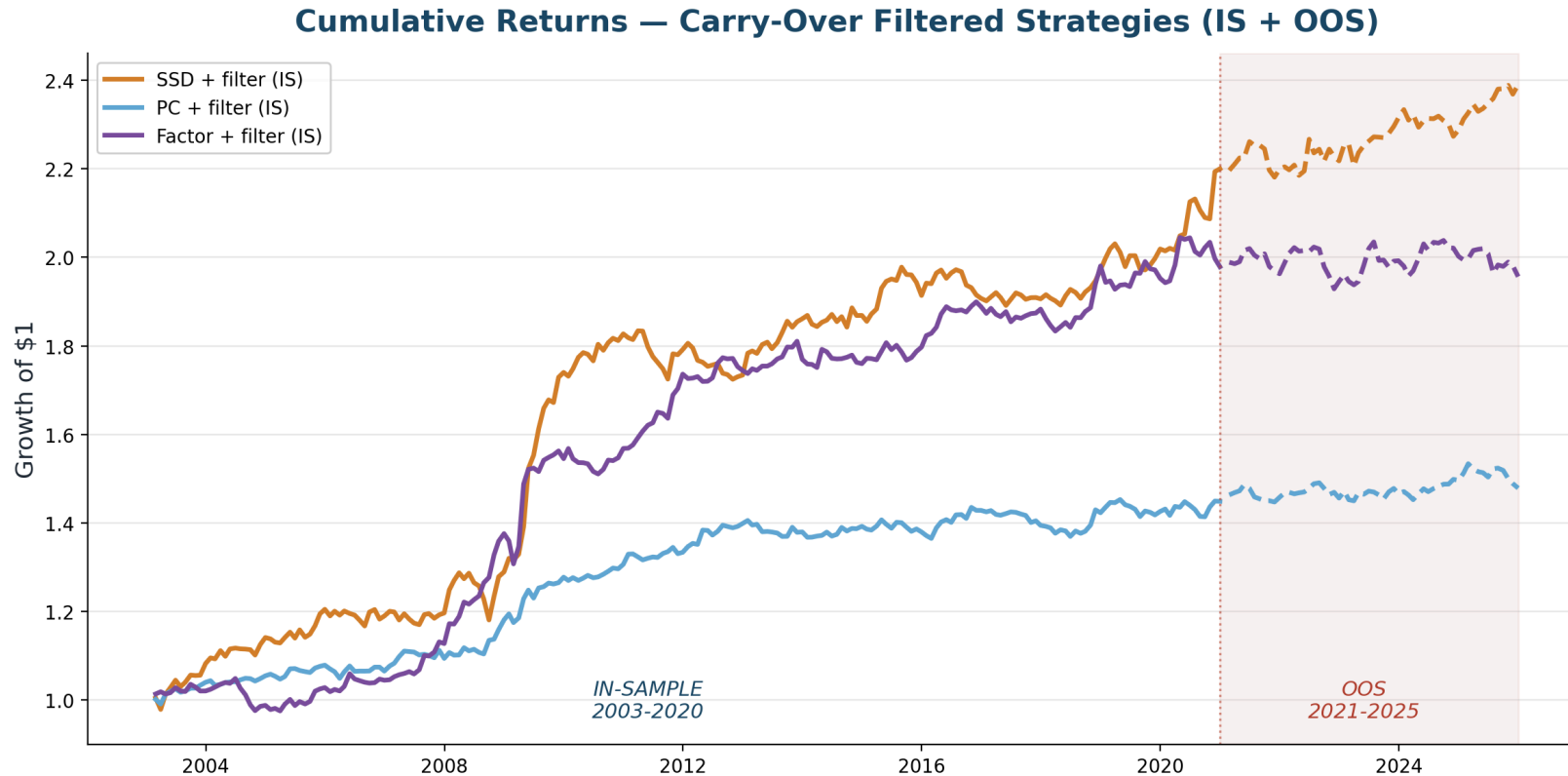
All results below use cointegration-filtered strategies with `carry_over=True`, `max_carry_months=3`.

Carry-over results — IS vs OOS scorecard

Strategy	Period	Ann. Ret	Ann. Vol	Sharpe	Max DD	Months
SSD + filter	IS 2003–2020	4.50%	5.10%	0.889	-8.27%	215
Factor + filter	IS 2003–2020	3.88%	4.54%	0.862	-7.00%	215
PC + filter	IS 2003–2020	2.09%	2.87%	0.736	-4.57%	215
SSD + filter	OOS 2021–2025	1.70%	3.21%	0.542	-3.55%	59
PC + filter	OOS 2021–2025	0.41%	2.41%	0.183	-3.60%	59
Factor + filter	OOS 2021–2025	-0.23%	3.22%	-0.056	-4.67%	59

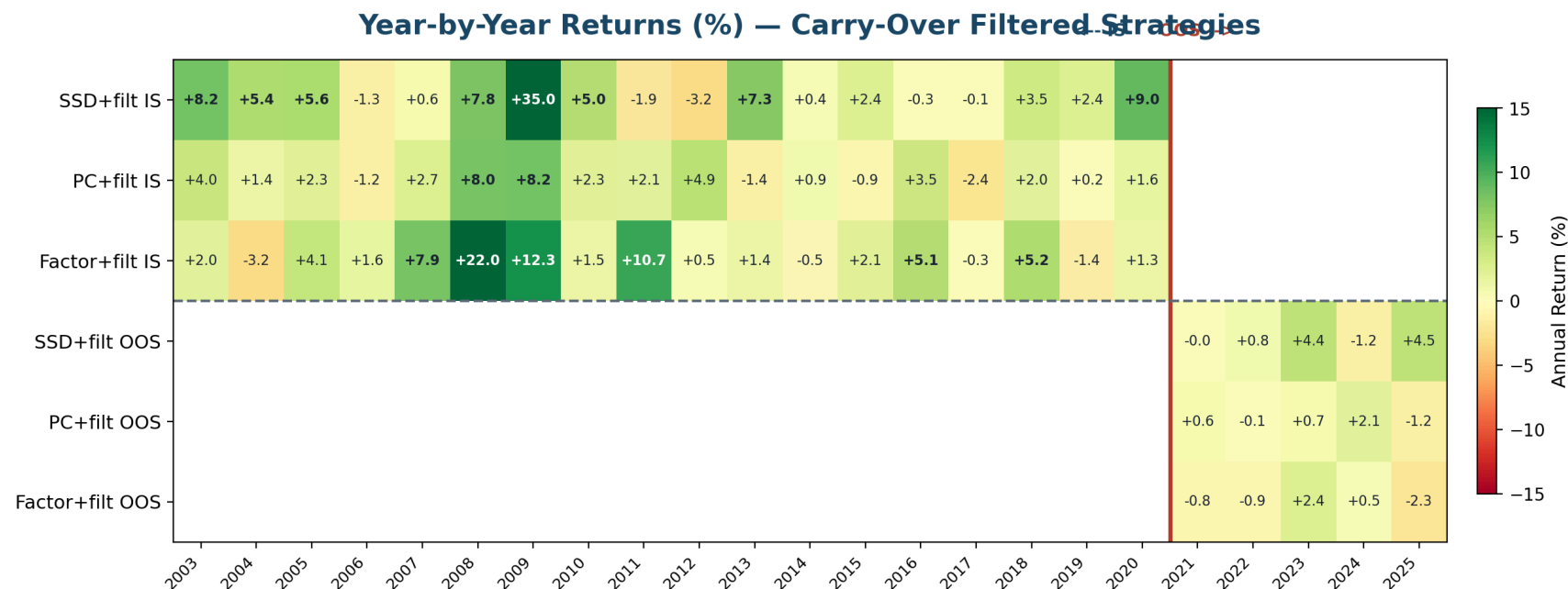
SSD + filter is the strongest carry-over strategy — 0.889 IS, 0.542 OOS.

Cumulative returns — carry-over strategies (IS + OOS)



Solid = IS (2003–2020), dashed = OOS (2021–2025). All strategies use `carry_over=True`, max 3 months.

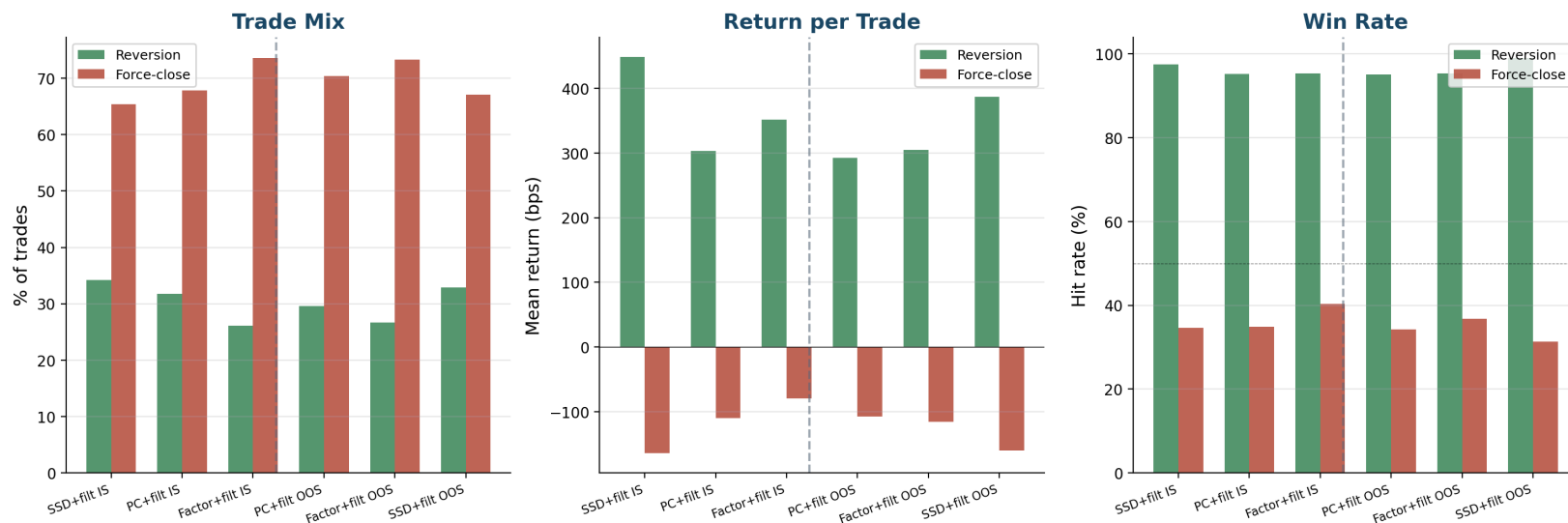
Year-by-year returns — carry-over strategies



- **SSD + filter:** consistently positive across years, strong in GFC and OOS
- More time for reversion smooths out year-to-year variance for SSD
- PC and factor show more negative years when trades are held longer

P&L breakdown — carry-over strategies (IS vs OOS)

P&L Breakdown — Carry-Over Filtered Strategies (IS vs OOS)



- **Higher reversion rate** — more trades complete their mean-reversion cycle
- **Fewer force-closes** — the drag that was killing profitability is reduced
- **Trade-off:** trades held longer carry more risk per position

Carry-over key findings

1. SSD + filter benefits most from carry-over

- IS Sharpe 0.889 | OOS Sharpe 0.542
- SSD selects by price-path shape — when these pairs diverge, they genuinely need more time to converge

2. PC + filter works better with force-close

- IS Sharpe 0.736 | OOS Sharpe 0.183
- PC pairs selected by idiosyncratic correlation — when they haven't reverted by month-end, the correlation structure may have shifted. Cutting losses early is better.

3. Factor-beta also prefers force-close

- IS Sharpe 0.862 | OOS Sharpe -0.056
- Factor loadings shift over time — holding amplifies the mismatch

Takeaway: the optimal trade duration depends on the distance metric. SSD pairs revert slowly (price convergence); PC/factor pairs revert quickly or not at all.

Engineering rigour

- **Modular package** — distances · clustering · spread · cointegration · backtest · performance · costs · lookahead, each independently unit-tested.
- **67 synthetic unit tests** across 11 files — validate every component on data with a *known* answer before touching real CRSP.
- **6/6 lookahead audit PASS** — black-box test confirms no future information leakage.
- **Look-ahead protection** baked into the z-score and the rolling formation/trading split.
- **Reproducible** — config-driven single source of truth; hyperparameters frozen pre-results to prevent overfitting.

Stack: Python · pandas · NumPy · scikit-learn (OPTICS) · statsmodels (ADF / OLS) · matplotlib · parquet.

What this demonstrates

- **Independent replication** of a 2025 research paper — read it, rebuilt it, matched the headline number on data we sourced ourselves.
- **Honest validation** — 5-year OOS test shows the strategy survives but with regime-dependent decay. We report what we found, not what we hoped.
- **Quant intuition** — diagnosed *why* the strategy makes money, then engineered the fix.
- **Discipline** — survivorship-bias-free data, out-of-sample design, frozen hyperparameters, unit-tested code, implementation-correctness review.

Thank you

Clustering-Based Pairs Trading

IS Sharpe **1.113** · OOS Sharpe **0.412** · 274 months total · built from the data up

Happy to walk through the code, the backtest engine, or the attribution analysis.